

Project #1 - CAD & 3D Printing

Gear-Driven Ferris Wheel Mechanism

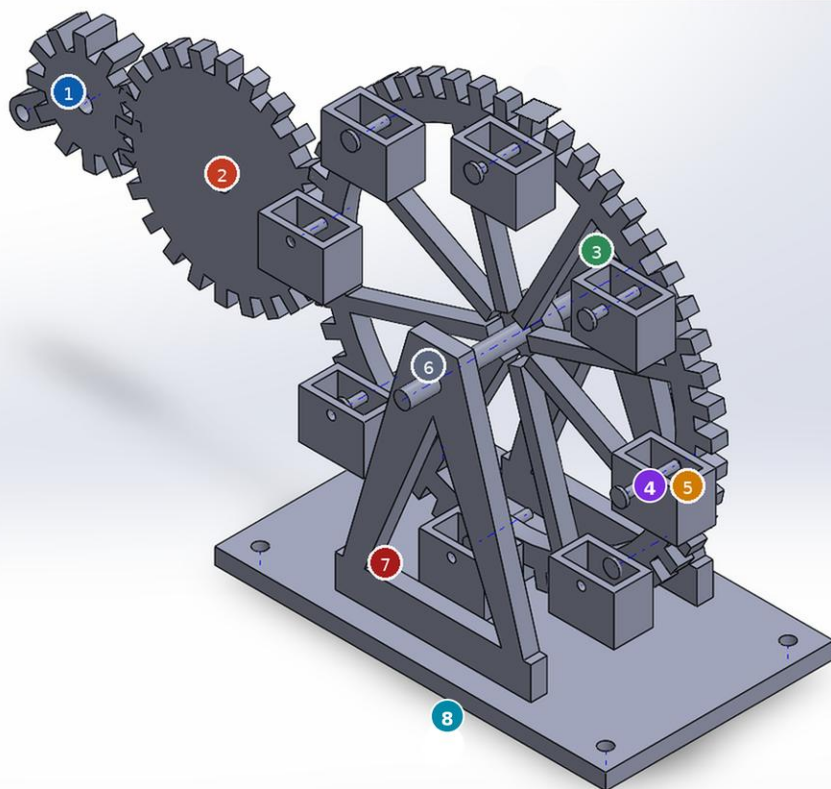
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Overview

I have loved Ferris wheels since I was a kid and wanted to use my SolidWorks abilities to create the best version possible with the skills I have developed. The simplified mechanism contains a 12-tooth input gear with an integrated crank, a 24-tooth idler gear, and a 48-tooth ring gear integrated into the Ferris wheel. Turning the crank rotates the input gear, which drives the idler gear, and the idler then drives the main wheel. The remaining components are the wheel spokes and hub, two A-frame supports, a base plate, the main axle shaft, eight gondolas, and eight headed pivot pins. The gear train provides the three dependent moving parts required by the project, while each gondola has an additional pivoting motion about its pin.

For this project, I created eight unique SolidWorks part files and assembled repeated instances where needed. I modeled the wheel and gears with extruded profiles, circular patterns, and cut features; created the axle with a revolved profile; created the gondola as a shelled rectangular body; and created a headed pivot pin with a chamfered insertion end. I then used coincident, concentric, distance, width, parallel, and gear mates to position the components and preserve their intended motion. The final model is a simplified tabletop demonstration of the gearing, support structure, wheel rotation, and freely pivoting gondola connection used in a Ferris wheel.



Major components

- 1 Input gear + crank
- 2 Idler gear
- 3 Ferris wheel / ring gear
- 4 Gondolas (8)
- 5 Gondola pivot pins
- 6 Main axle shaft
- 7 A-frame supports (2)
- 8 Base plate

Figure 1. Overall SolidWorks assembly of the gear-driven Ferris wheel with the major components labeled.

How It's Made

A full-size Ferris wheel is primarily a fabricated structural-steel assembly. The rim, spokes, hub, and support frame are built from steel plate, tube, and rolled structural sections because these materials provide high stiffness, fatigue resistance, weldability, and durability outdoors. The passenger gondolas are commonly formed or welded from sheet metal and supported by pivot joints so gravity keeps them upright as the wheel turns. The drive train uses machined steel gears, shafts, bearings, and braking components because these parts must transmit torque while maintaining controlled alignment and predictable motion. My SolidWorks design simplifies these systems into plate-like parts while preserving the same functional relationships among the crank, idler gear, main wheel, support frame, and gondola pivots.

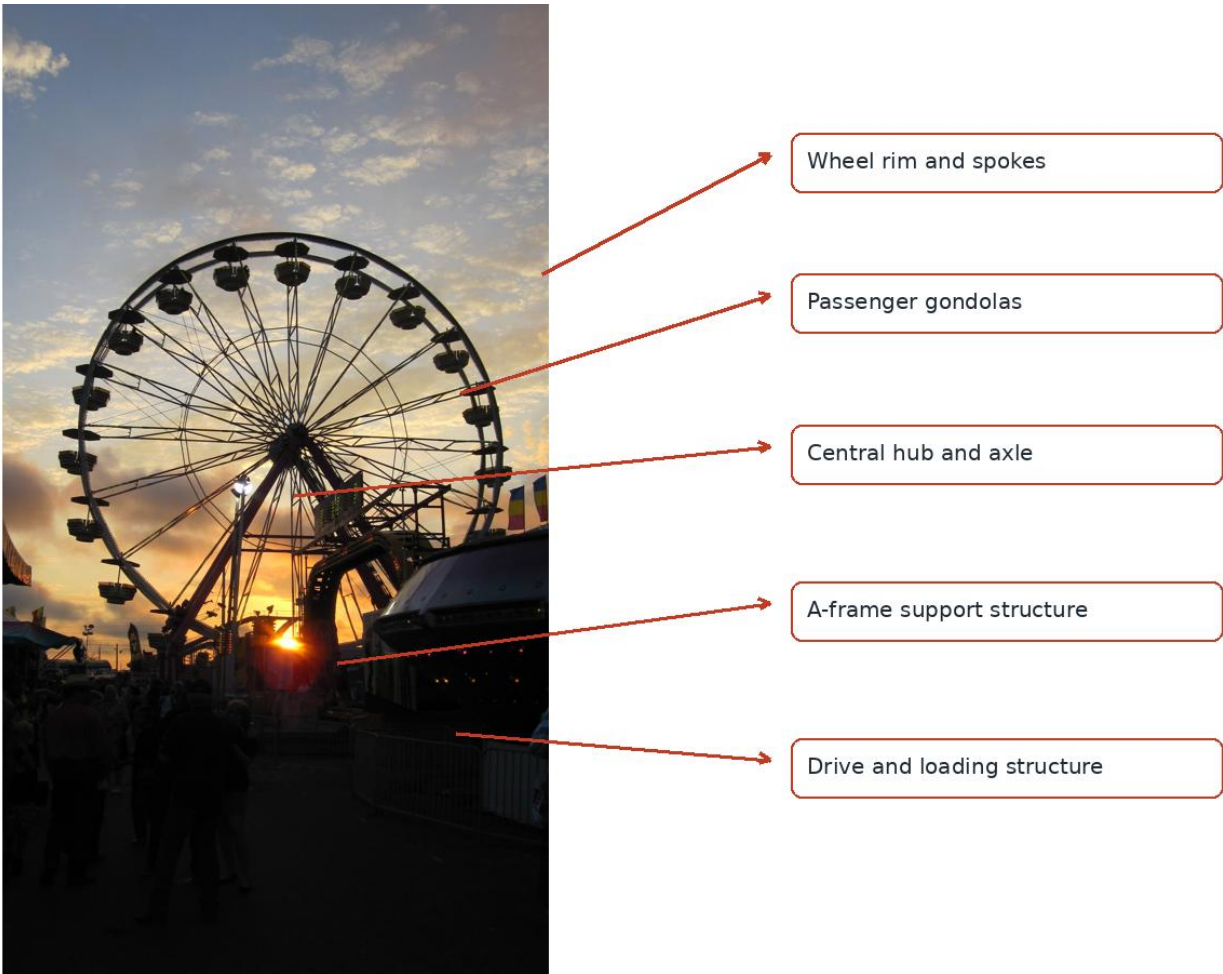


Figure 2. A real Ferris wheel showing the wheel structure, passenger gondolas, central hub, and support frame.

Table 1. Likely conventional manufacturing routes for three major Ferris-wheel components.

Conventional Mfg Steps	Main wheel / ring gear	Input and idler gears	Gondola carriage
Raw Stock	Structural-steel plate, tube, and rolled sections	Alloy-steel plate or round-bar blanks	Low-carbon steel or aluminum sheet
Process 1	Cut hub plates, spoke members, rim segments, and gear-ring sections to rough size	Saw blanks to length and face both sides	Blank the wall and floor geometry from sheet stock
Process 2	Roll rim sections to the required radius and fixture the wheel geometry	CNC turn the outside diameter, faces, and center bore	Brake-form or deep-draw the body into a box-like carriage

Conventional Mfg Steps	Main wheel / ring gear	Input and idler gears	Gondola carriage
Process 3	Weld or bolt spokes between the hub and rim; attach ring-gear segments	Hob, shape, or CNC mill the gear teeth	Weld, rivet, or stake the corners and reinforcement features
Process 4	Drill and ream the axle bore and gondola mounting holes	Drill the crank-pin feature in the input gear and ream the center bores	Drill and ream coaxial pivot holes in the side walls
Process 5	Stress-relieve as needed, inspect runout, and deburr all edges	Heat treat for strength and tooth wear resistance	Deburr edges and inspect the pivot-hole alignment
Process 6	Blast, prime, and powder-coat or paint for corrosion protection	Finish-grind critical faces/bores and apply protective coating	Powder-coat or paint; install seating and safety hardware
Finished Product	Aligned, coated wheel structure with integrated drive ring	Dimensionally controlled drive gears ready for shafts/bearings	Finished passenger carriage that pivots about its support pin

The main wheel is suited to structural fabrication because its diameter is large relative to the thickness of its members, and the design must carry cyclic bending and passenger loads economically. Rolled and welded steel sections provide a better strength-to-cost balance than machining the entire wheel from a single block. The gears require accurate tooth spacing, bore alignment, and wear resistance, so machining followed by heat treatment is appropriate. The gondola is a thin-walled enclosure, which makes sheet-metal forming and joining more economical than machining. In all three cases, corrosion-resistant finishing is important because an actual amusement-ride assembly operates outdoors and is repeatedly exposed to moisture and handling.

Part Models

Modeling Approach

The assembly was divided into separate part files so each component could be modeled with the operation that best matched its geometry. Table 2 summarizes the main feature strategy and the principal dimensions used in the completed model.

Table 2. SolidWorks modeling approach for the individual parts.

Part	Modeling approach	Key design information
Base plate	Create a centered rectangular sketch, extrude it to thickness, and cut four mounting holes through the plate.	6.00 × 3.50 × 0.25 in
A-frame support	Extrude the lower foot and tapered outer frame, cut the triangular opening, and cut the axle hole near the top. Use the same part twice in the assembly.	Approx. 3.00 in wide × 4.00 in high × 0.25 in thick
Axle shaft	Sketch a half-section with a construction centerline and revolve it through 360 degrees.	Ø0.25 in × 3.75 in
Ferris wheel / ring gear	Extrude concentric rim and hub regions, create one spoke and circular-pattern it, create one tooth and pattern it 48 times, and cut the center and eight gondola holes.	Ø5.45 in outer diameter; 48 teeth; 8 gondola locations
Idler gear	Extrude the gear blank, extrude one tooth, circular-pattern the tooth 24 times, and cut the center bore.	24 teeth; Ø0.25-in center bore
Input gear + crank	Build the 12-tooth gear from an extruded disk and tooth pattern, add the crank arm as an extruded boss, and cut the gear and handle holes.	12 teeth; integrated crank arm
Gondola	Extrude a centered rectangular block, remove the top face with a 0.10-in shell, and cut a coaxial pivot hole through both side walls.	1.00 × 0.60 × 0.75 in; 0.10-in walls; Ø0.13-in holes
Gondola pivot pin	Create a Ø0.125-in shank, add a Ø0.25-in retaining head, and chamfer the free end.	0.85-in shank; 0.05-in head thickness; 0.01-in chamfer

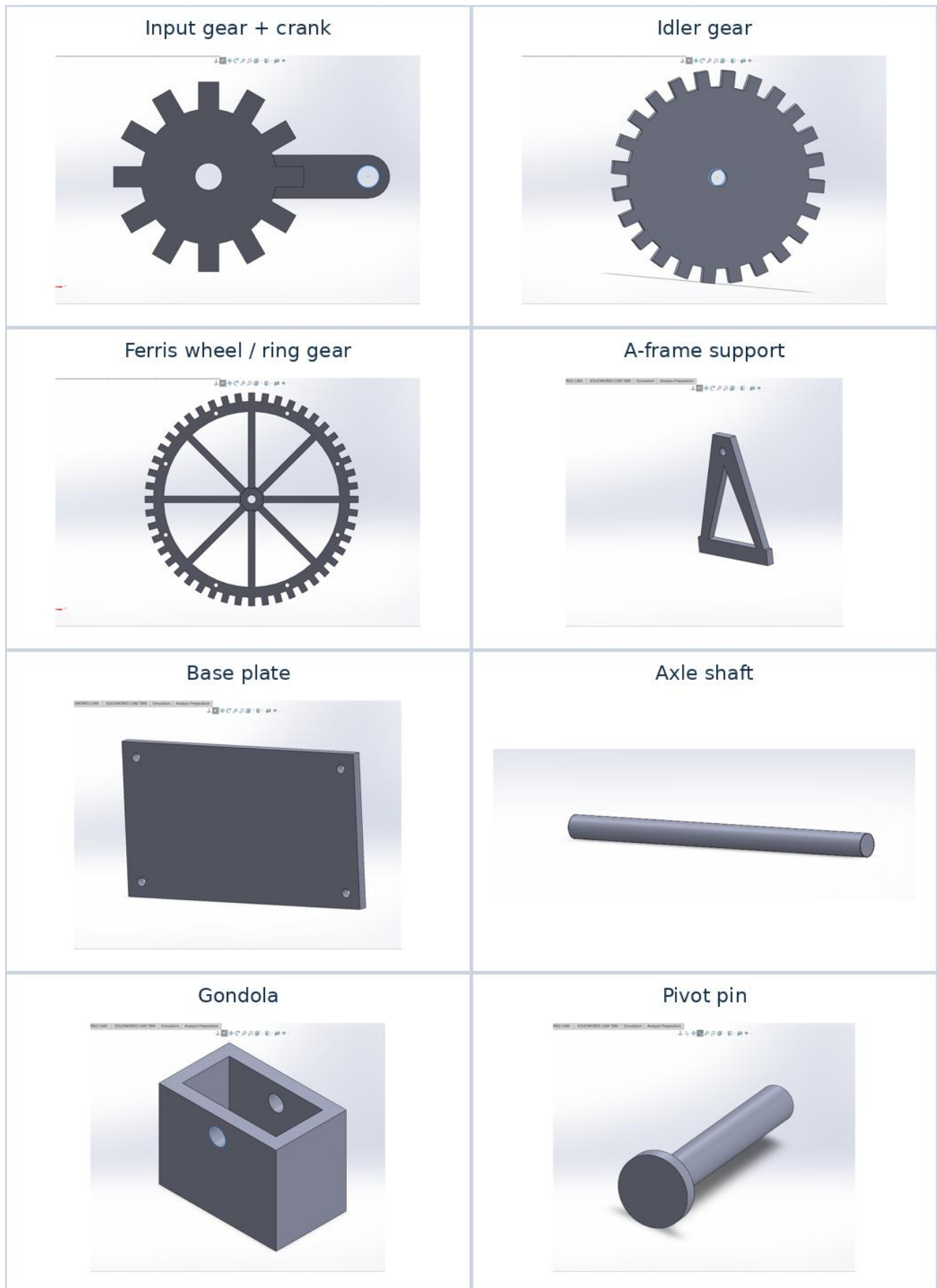


Figure 3. Major SolidWorks part models used in the Ferris-wheel assembly.

CAD Operations

The completed files use the CAD operations best suited to each component, including extrusion, revolution, shelling, circular patterning, cutting, and chamfering. Table 3 summarizes the operations and SolidWorks features used for every unique part, while Figure 4 shows representative extrusion and revolution examples.

Table 3. CAD operations and corresponding SolidWorks features for all unique parts.

Part name	CAD operation(s)	SolidWorks feature(s)	Purpose in the model
Base plate	Extruded Boss/Base; Extruded Cut	Boss-Extrude1; Cut-Extrude1	Extrudes the rectangular plate to thickness and cuts the four mounting holes.
A-frame support	Extruded Boss/Base; Extruded Cut	Boss-Extrude1; Boss-Extrude2; Cut-Extrude1; Cut-Extrude2	Builds the lower foot and tapered frame, then removes the triangular opening and axle hole.
Axle shaft	Revolved Boss/Base	Revolve1	Revolves a half-profile through 360° about a construction centerline to create the cylindrical axle.
Ferris wheel / ring gear	Extruded Boss/Base; Circular Pattern; Extruded Cut	Boss-Extrude features; CirPattern1, CirPattern2, and CirPattern6; Cut-Extrude features	Creates the rim, hub, spokes, and one gear tooth; patterns the repeated geometry; cuts the axle and gondola holes.
Idler gear	Extruded Boss/Base; Circular Pattern; Extruded Cut	Boss-Extrude features; circular tooth pattern; Cut-Extrude1	Creates the gear blank and one tooth, repeats the tooth 24 times, and cuts the center bore.
Input gear + crank	Extruded Boss/Base; Circular Pattern; Extruded Cut	Boss-Extrude features; circular tooth pattern; Cut-Extrude features	Creates the 12-tooth gear, adds the crank arm, and cuts the gear bore and handle hole.
Gondola	Extruded Boss/Base; Shell; Extruded Cut	Boss-Extrude1; Shell1; Cut-Extrude1	Extrudes the carriage block, shells it to form an open-top body, and cuts coaxial pivot holes through both walls.
Gondola pivot pin	Extruded Boss/Base; Chamfer	Boss-Extrude1; Boss-Extrude2; Chamfer1	Creates the cylindrical shank and retaining head, then chamfers the insertion end.

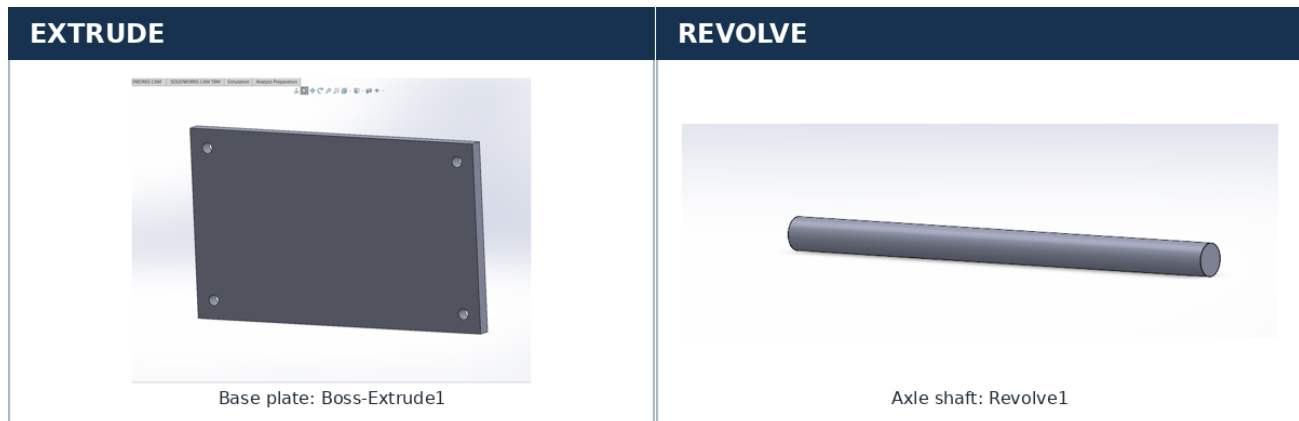


Figure 4. Extrusion and revolution examples used in the completed SolidWorks models.

Assembly Model

The assembly was built from the base upward so the support geometry established a stable reference for the moving components. The base plate was located with coincident mates to the assembly reference planes rather than using Fix. Each A-frame support was seated against the base with coincident faces and positioned using reference-plane, parallel, and distance relationships so that the two axle holes were coaxial. The main axle was then made concentric with the support holes and centered between the supports. The Ferris wheel was made concentric with the axle and centered across the support spacing, leaving only rotation about the axle as its intended degree of freedom.

The idler and input gears were positioned with their axes parallel to the main axle and with fixed center distances to the assembly planes. A gear mate connected the 24-tooth idler to the 48-tooth Ferris wheel, and a second gear mate connected the 12-tooth input gear to the 24-tooth idler. The gondola pivot holes were aligned with the wheel mounting holes, and each headed pivot pin was made concentric with the aligned holes. A distance mate placed the gondola outside the wheel, while a coincident mate seated the pin head against the gondola wall. This arrangement prevents axial sliding while allowing the gondola to rotate about the pin. Repeated gondola and pin pairs were created at the eight patterned wheel holes.

Table 4. Assembly constraints and intended degrees of freedom.

Component	Primary constraints	Intended motion
Base plate	Three coincident mates to the assembly Front, Top, and Right planes	No motion
A-frame supports	Bottom face coincident with base; planes/faces located by coincident, parallel, and distance mates	No motion
Main axle shaft	Concentric with both support holes; centered with a width mate	No translation; acts as wheel axis
Ferris wheel	Concentric with axle; centered between supports	One rotational degree of freedom about axle
Idler gear	Axis positioned with distance/reference-plane mates; gear mate to wheel	One rotational degree of freedom controlled by the wheel gear mate
Input gear + crank	Axis positioned with distance/reference-plane mates; gear mate to idler	One rotational degree of freedom; user input at crank
Gondola and pivot pin	Concentric hole/pin mates; axial distance and pin-head face mate	One rotational degree of freedom about the pivot pin

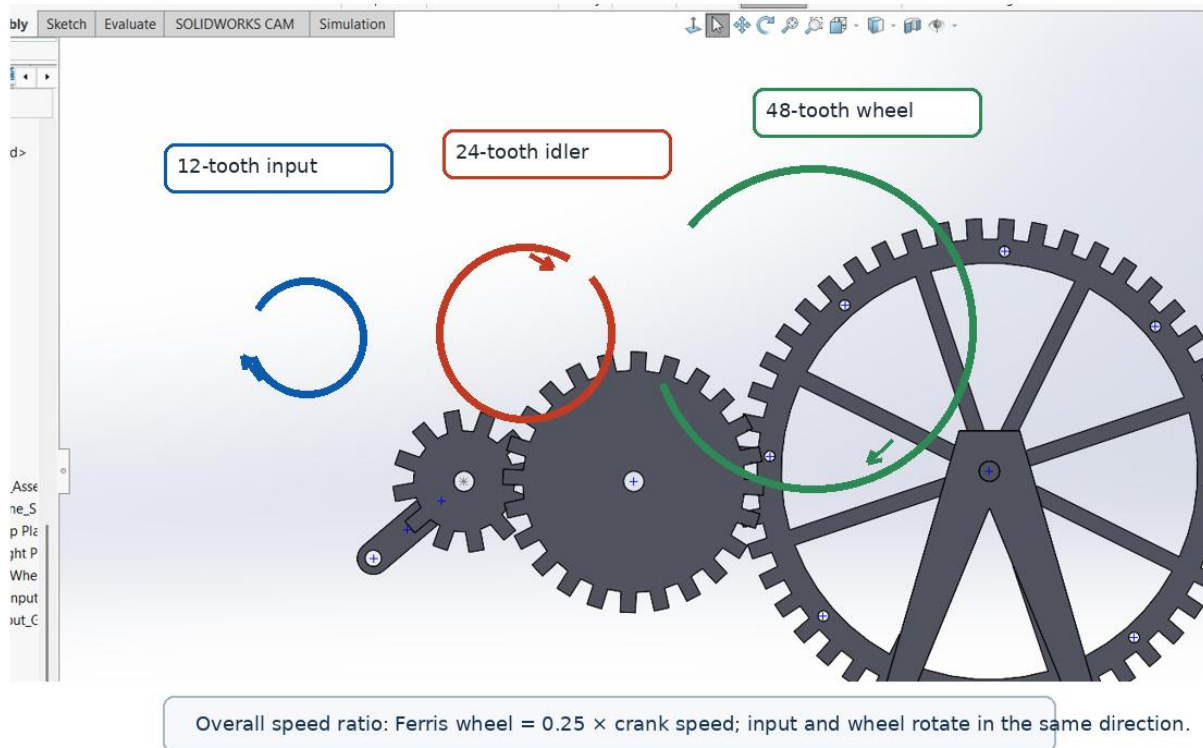


Figure 5. Front view of the gear train and the resulting 4:1 speed reduction from the crank to the Ferris wheel.

Partial exploded arrangement

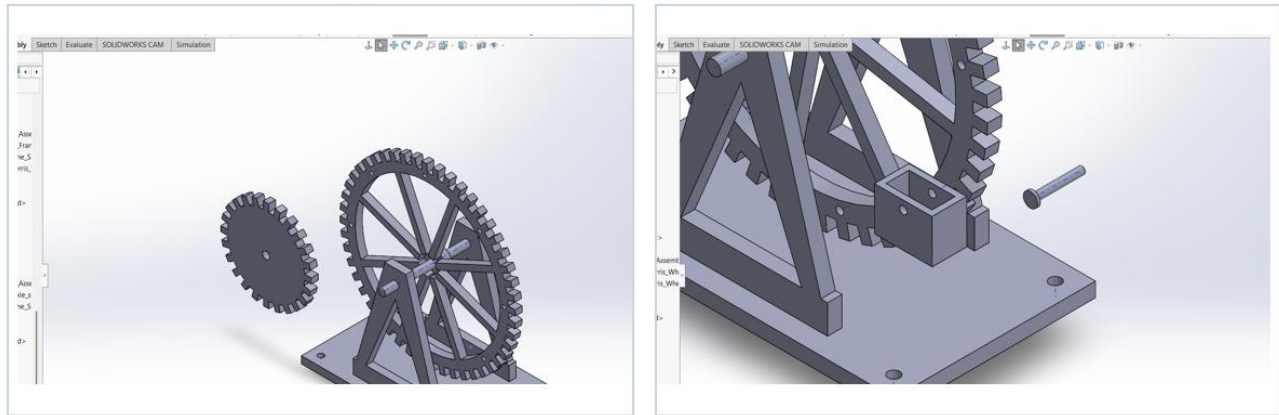


Figure 6. Partial exploded arrangement captured during assembly development.

Conclusion

This project produced a complete CAD concept for a gear-driven Ferris wheel mechanism. The model contains a hand crank, two-stage external gear train, spoked wheel with an integrated ring gear, support frame, axle, and repeated pivoting gondolas. The 12-tooth input gear, 24-tooth idler, and 48-tooth wheel create a 4:1 reduction, so the large wheel turns more slowly and smoothly than the crank. The assembly work demonstrated how mates can fully locate components while intentionally retaining only the rotational degrees of freedom needed for operation.

The most important lessons were planning the order of features, using circular patterns for repeated geometry, selecting stable assembly references, and distinguishing between a part that is visually aligned and a part that is correctly constrained. The gondola connection also showed the importance of adding a separate pivot pin and controlling axial clearance instead of allowing the wheel to intersect the carriage.